

Geometric Sequences and Series



Geometric sequences

- 1 Consider the sequence 3, 6, 12, 24, ...
 - a State the common ratio.
 - b Write a formula for the n th term.
 - c Find a_{10} .
 - 2 In a geometric sequence, $a_2 = 6$ and $a_5 = 48$. Find the common ratio and the first term.
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Geometric series

- 3 Find the sum of the first 8 terms of $5 + 10 + 20 + \dots$
- 4 Evaluate each infinite sum, or state that it diverges:
 - a $18 + 6 + 2 + \frac{2}{3} + \dots$
 - b $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots$
 - c $2 + 3 + 4.5 + \dots$
- 5 Write the repeating decimal $0.\overline{7}$ as a fraction using an infinite geometric series.
- 6 A ball is dropped from 8 m and rebounds to $\frac{3}{4}$ of its previous height on each bounce. Find the total vertical distance it travels (falling and rising) before coming to rest.